

Kiyan Amirian

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Research Summary

PhD student in Materials Science and Engineering at the University of Maryland, working at the intersection of machine learning and computational materials science. Research focuses on diagnosing where machine-learned interatomic potentials fail in realistic atomistic simulations, spanning force prediction, phase stability, and ion migration in solid-state ionic conductors, to guide the development of more reliable models.

Education

University of Maryland

Doctor of Philosophy in Materials Science and Engineering

Cumulative GPA: 4.0/4.0

College Park, MD, USA

2023–Present

Shiraz University

Bachelor of Science in Materials Science and Engineering

Cumulative GPA: 17.15/20 (3.5/4.0); Last 2-year GPA: 18.54/20 (3.9/4.0)

Shiraz, Fars, Iran

2018–2023

Research Interests

Computational Materials Science

Atomistic Simulation

Machine-Learned Interatomic Potentials

AI for Science

Publications

K. Amirian, RS Saravanan, F Adams, CE Schwarz, Y Mo

arXiv, 2026

FPBench: Application-Oriented Error Decomposition for Foundation Potentials

[arXiv:2609.05714](#) · [GitHub](#) · [Leaderboard](#)

K. Amirian, Z. Abbasi, R. Ebrahimi

2023

Analytical and numerical instability analysis of functionally graded low-carbon steel.

Steel Symposium 401, International Journal of Iron and Steel Society of Iran.

[doi:10.22034/IJISSI.2023.1990580.1262](#)

Research Experience

Graduate Researcher, UMD (Advisor: Prof. Yifei Mo)

2024–Present

- Developed FP-DeErr, an application-oriented error-decomposition framework for foundation potentials (FPs) that resolves errors by the physical quantities, configurations, and workflow stages governing each computational task; benchmarked multiple FPs against DFT using ASE, MatCalc, and pymatgen on HPC (SLURM) ([GitHub](#), [leaderboard](#)).
- Force prediction: decomposed force errors into magnitude and angle, and designed metrics for highly accurate predictions, large-force-error atoms, and far-from-equilibrium atoms (MatPES, OMat24), showing that these failure modes are diluted in average MAE/RMSE ([GitHub](#), [leaderboard](#)).
- Phase stability and elemental ordering: designed convex-hull agreement metrics (ground state, within-phase and global hull minimum) and evaluated ordering rankings (Top-1, Recall@k, Spearman's ρ) on chalcogenide phase-change materials, separating intrinsic energy-ranking errors from FP-relaxation errors ([GitHub](#), [leaderboard](#)).
- Ion migration: benchmarked FP-NEB against DFT-NEB on 154 Li- and Na-ion pathways with metrics for convergence, profile shape, endpoint energy ranking, and barriers, tracing barrier errors to intrinsic PES error, endpoint relaxation, and along-path forces ([GitHub](#), [leaderboard](#)).

Graduate Researcher, UMD (Advisor: Prof. Ji-Cheng 'JC' Zhao)

Fall 2023

- 3D printing of High Entropy Alloys.

Undergraduate Researcher, Shiraz University (Advisor: Prof. Ramin Ebrahimi)2019–2022

- Numerical analysis on instability of Functionally Graded Materials (FGM) ([paper](#)).

Selected Projects

Image Classification: Classical ML and Transfer LearningSpring 2025

Built pipelines for MNIST and a 10-class species dataset using logistic regression from scratch, RBF SVM+PCA on MNIST (97.1%), and VGG19 transfer learning with a frozen backbone.

Face Recognition: Dimensionality Reduction and Kernel ClassifiersSpring 2025

Benchmarked PCA/MDA with Bayes, k-NN, SVM, and AdaBoost on a 200-subject dataset. Implemented a custom gradient-descent kernel SVM to address CVXOPT numerical failures with high-degree polynomial kernels.

KV Cache Optimization for Large Language ModelsFall 2025

Led a team comparing KV-cache compression techniques (StreamingLLM, H2O, KIVI, Zip-Cache) for long-context LLM inference. Reimplemented KIVI quantization and designed evaluation workflows across summarization, reasoning, and QA benchmarks.

Powder Bed Fusion Thermal SimulationSpring 2022

Implemented custom FORTRAN subroutines in ABAQUS to model laser–material interaction including spatially varying heat flux and temperature-dependent material properties.

Skills

Programming: Python (NumPy, Matplotlib), Bash, MATLAB, Fortran, Git, $\LaTeX$
Atomistic Simulation: DFT (VASP), LAMMPS, ASE, pymatgen, matcalc; NEB, molecular dynamics
Foundation Potentials: 20+ FPs benchmarked against DFT, including MACE, CHGNet, M3GNet, UMA, Orb, MatterSim, DPA4, and eSEN
ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn
ML Methods: Transfer learning, convolutional neural networks, support vector machines
HPC: SLURM job arrays on university HPC clusters
Continuum Modeling: Abaqus (FEA)

Honors & Awards

Department of Materials Science and Engineering, University of MarylandCollege Park, MD

University of Maryland Dean’s Fellowship Award, \$43,000Spring 2023

Department of Materials Science and Technology, Shiraz UniversityShiraz, Fars, Iran

Ranked 1st among 56 students in B.Sc. Materials Science and EngineeringSpring 2022

Relevant Graduate Courses

Introduction to Atomistic Modeling (4.0/4.0)

Natural Language Processing (4.0/4.0)

Statistical Pattern Recognition (4.0/4.0)

Machine Learning for Materials Science (4.0/4.0)

Teaching Experience

Kinetics, Diffusion and Phase Transformations (Fall 2024)

Mechanical Properties II (Spring 2022)

Fundamentals of Computer (Spring 2020)

Algebra (Fall 2019)

References

Yifei Mo

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